Privacy-Preserving Fall Detection System

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Falls: a Public Health Challenge

1/3

Adults aged 65+ fall each year

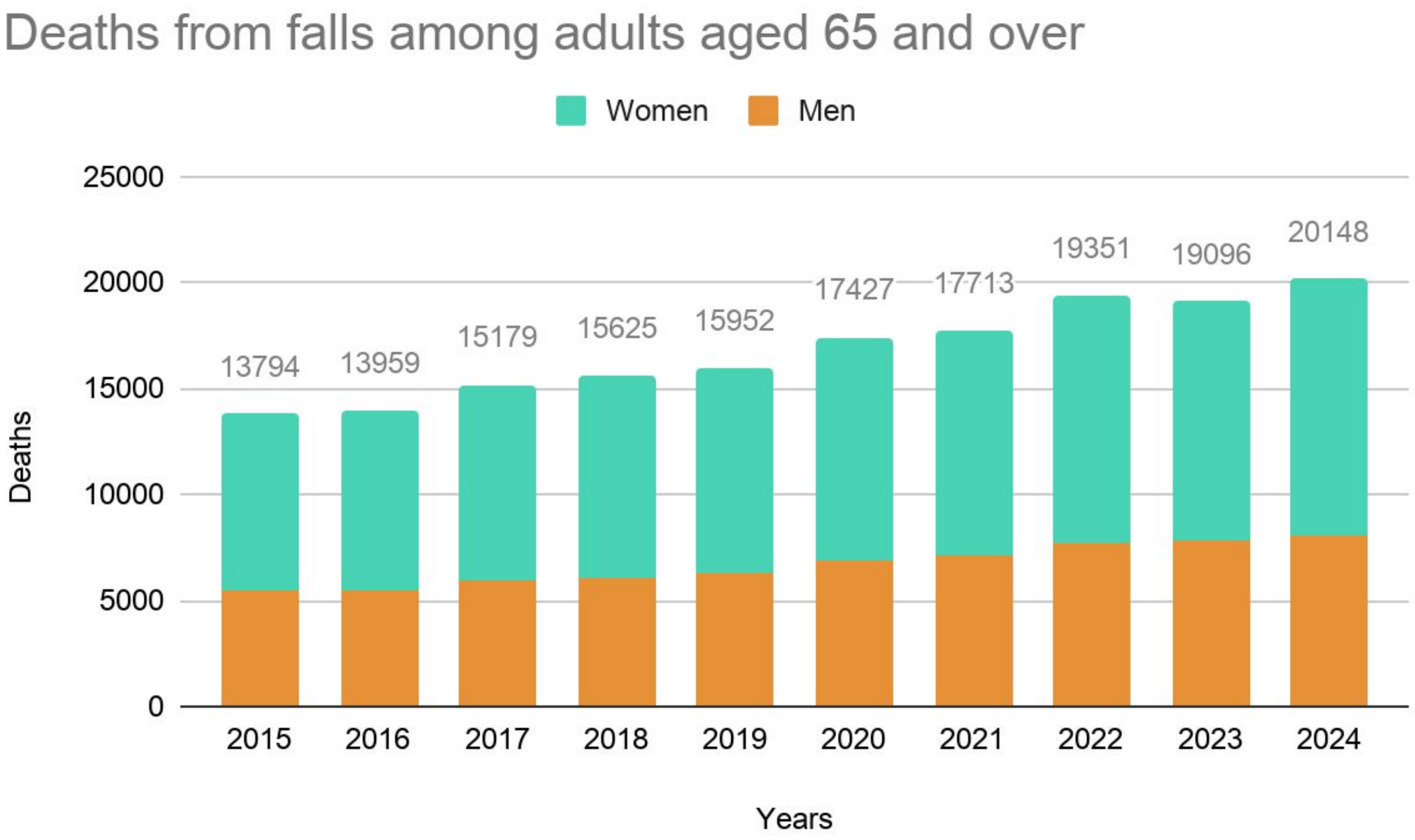
20k

Deaths from falls, France, 2024 (+18%)

Leading cause of accidental death among older adults

Source: Santé Publique France https://www.santepubliquefrance.fr/chute/donnees

A Growing Issue



Source: Santé Publique France https://www.santepubliquefrance.fr/chute/donnees

The "Long Lie" Fall – A Critical Emergency

- **Delayed Rescue:** Lying on the floor for more than an hour drastically increases complications.
- **Consequences :** Dehydration, muscle and tissue damage, hypothermia, delayed medical treatment, psychological trauma
- **The 6-months toll :** **50% mortality rate within 6 months** following a “Long Lie” fall (>1h) for adults over 65, even without direct injuries from the fall

Source: **Physiopedia** https://www.physio-pedia.com/Long_Lie

The Privacy Concern

Cameras Opt-Out

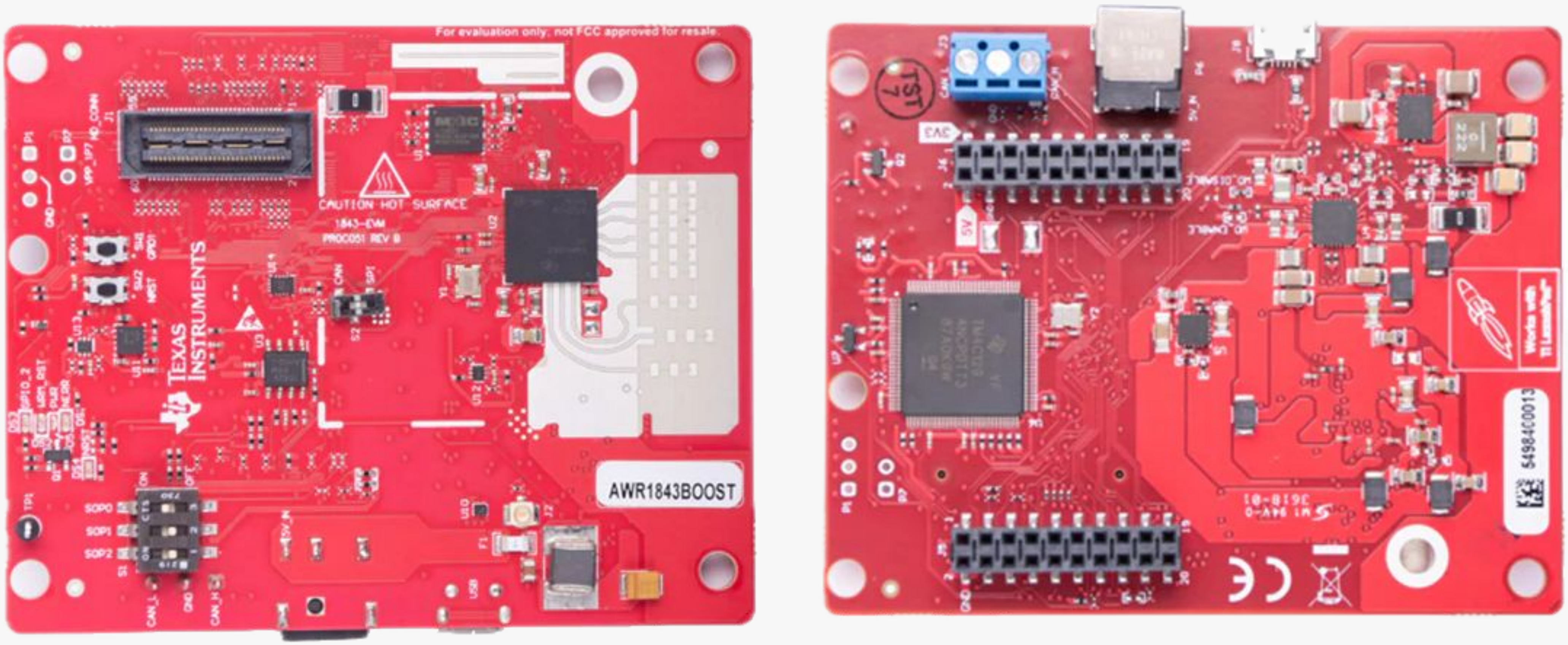
80% to 90% of older adults refuse optical cameras in their bedroom or bathroom.

High-Risk Area

Bathroom is the #1 location for falls, yet it is also the most private space.

Source: Kan J., SFU, 2025 https://summit.sfu.ca/_flysystem/fedora/2025-07/etd21193.pdf

Solution: mmWave Radars

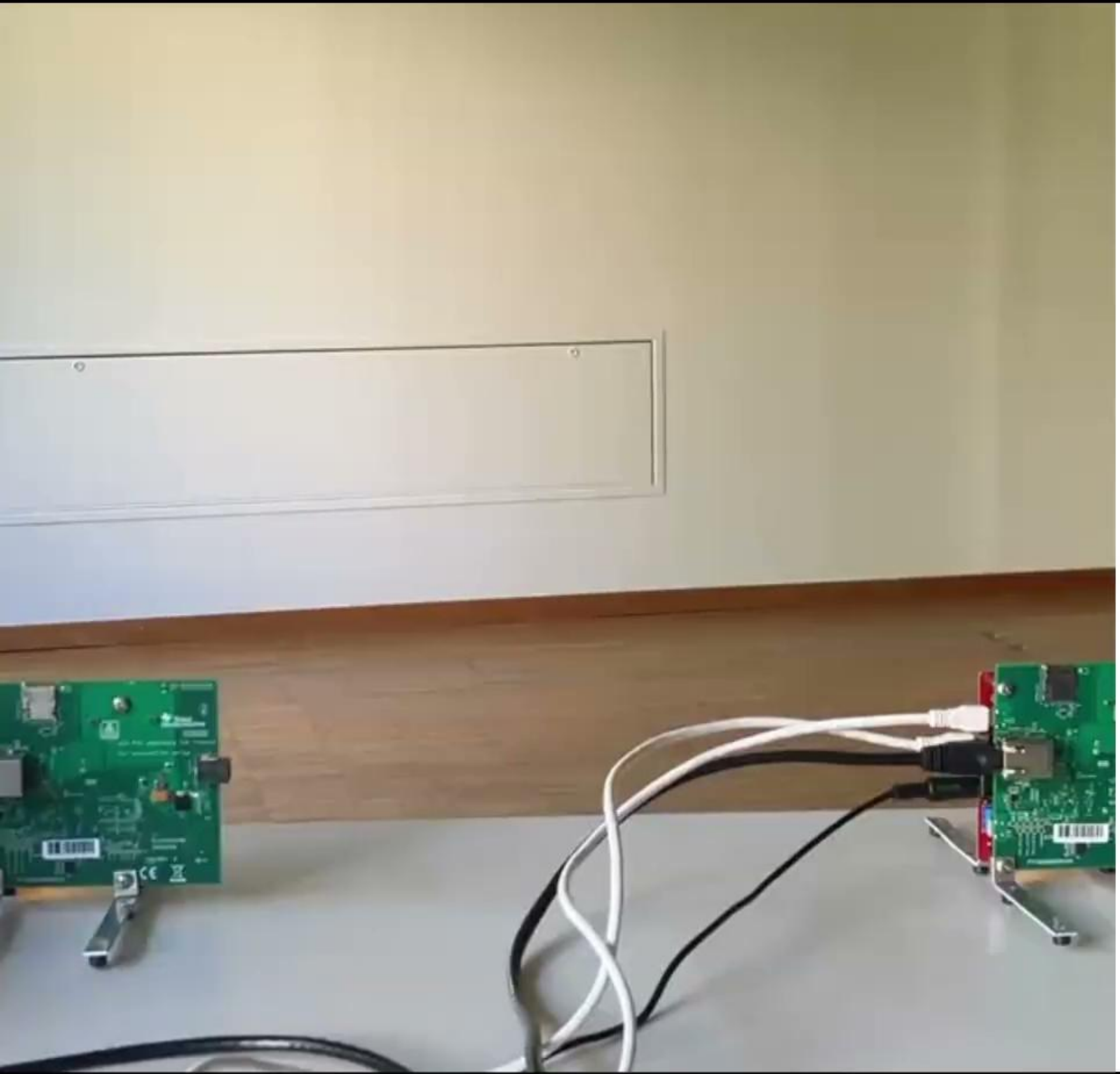


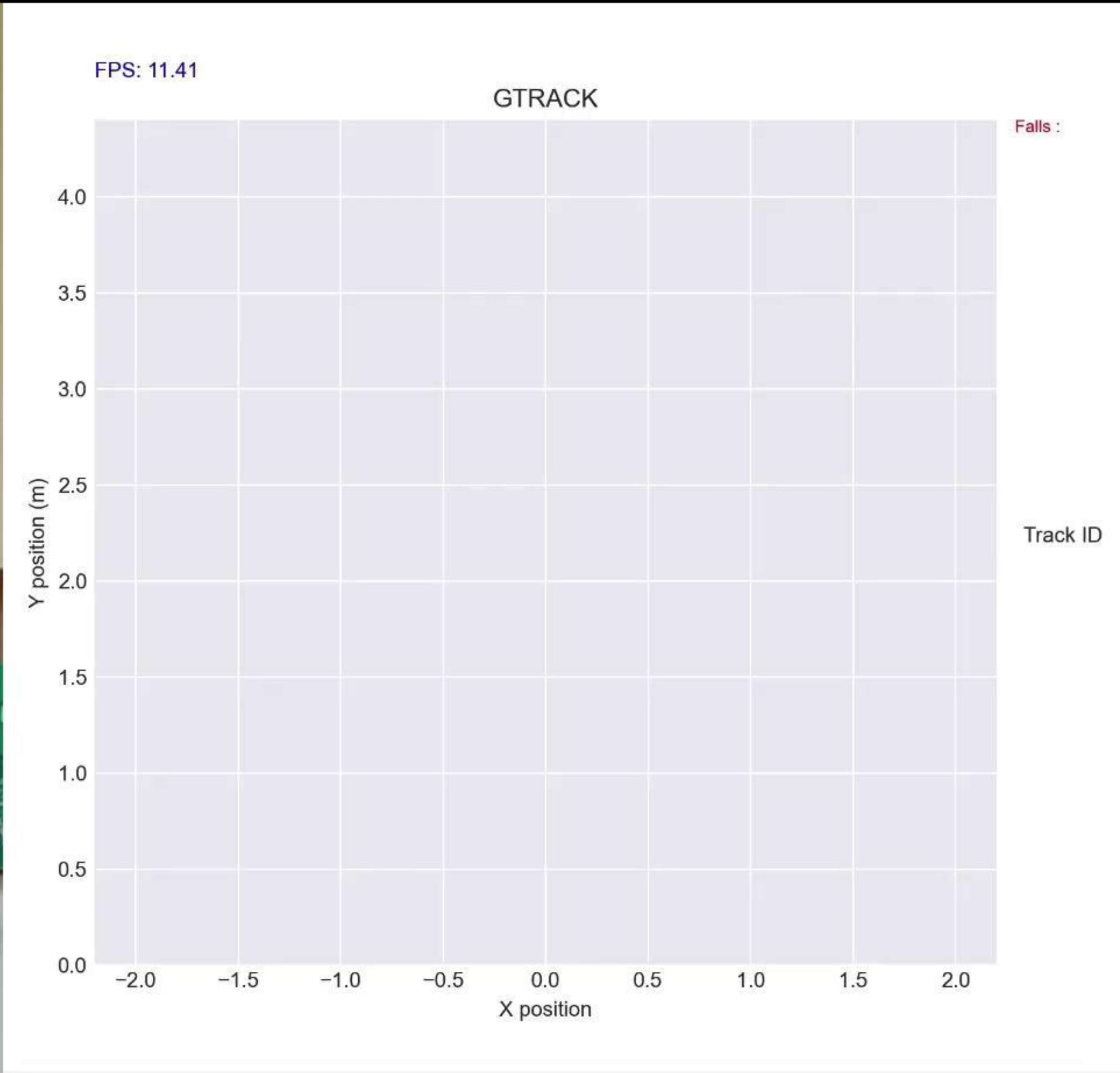
Technologies Comparison

Metric	Optical Cameras	Wearables (Bracelets)	Radar mmWave
Privacy Preservation	❌ Non	✅ High	⭐ **Superior (Anonymized)**
Auto-Detection	✅ Yes	❌ Requires Manual Action	✅ **Fully Automatic**
Low Light / Bathroom	❌ Limited	✅ Yes	✅ **Yes (RF Sensing)**
User Compliance	❌ Low (Privacy resistance)	⚠ Medium (Often forgotten)	✅ **High (Passive sensing)**

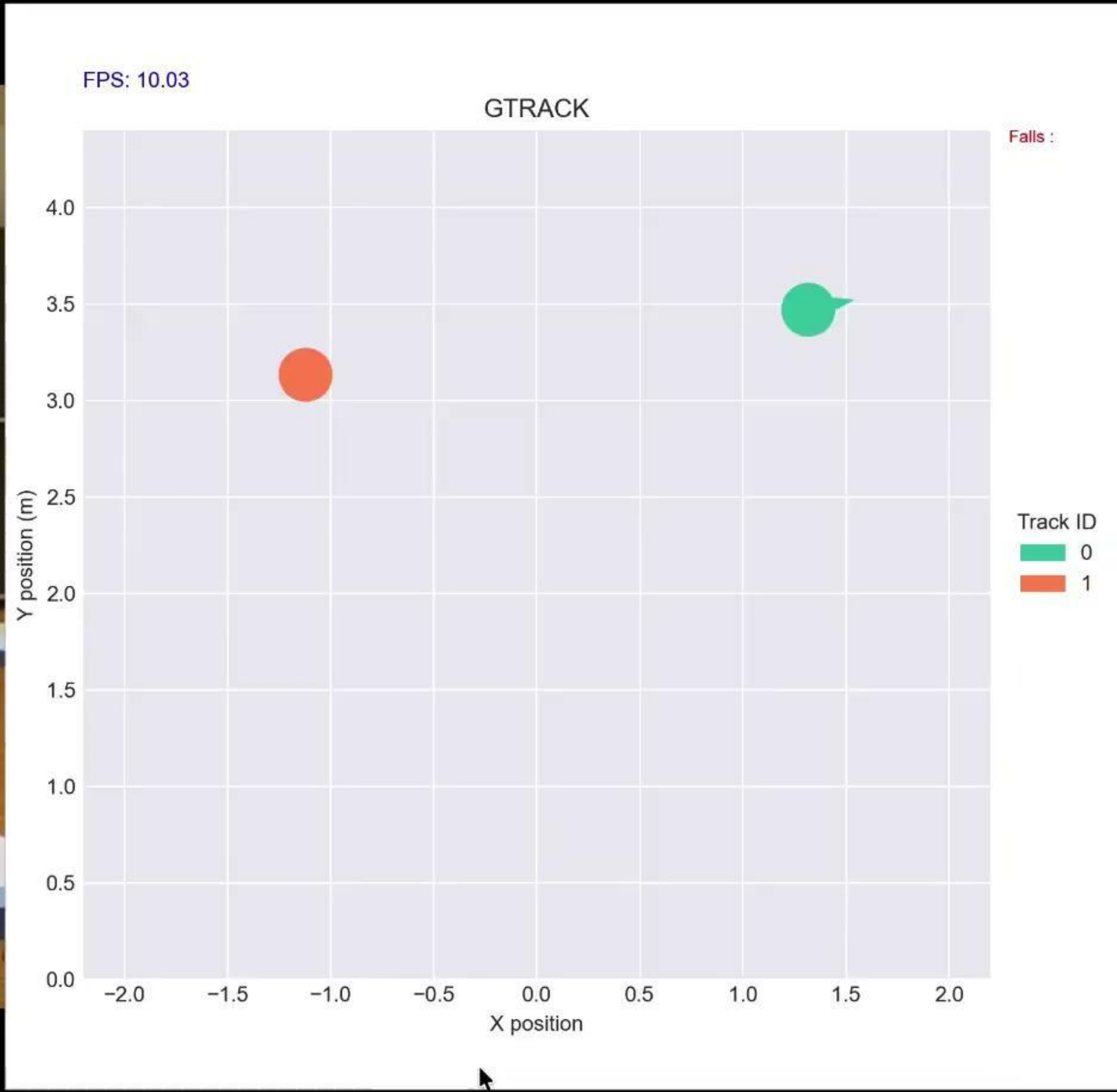
Multitracking

Presentation

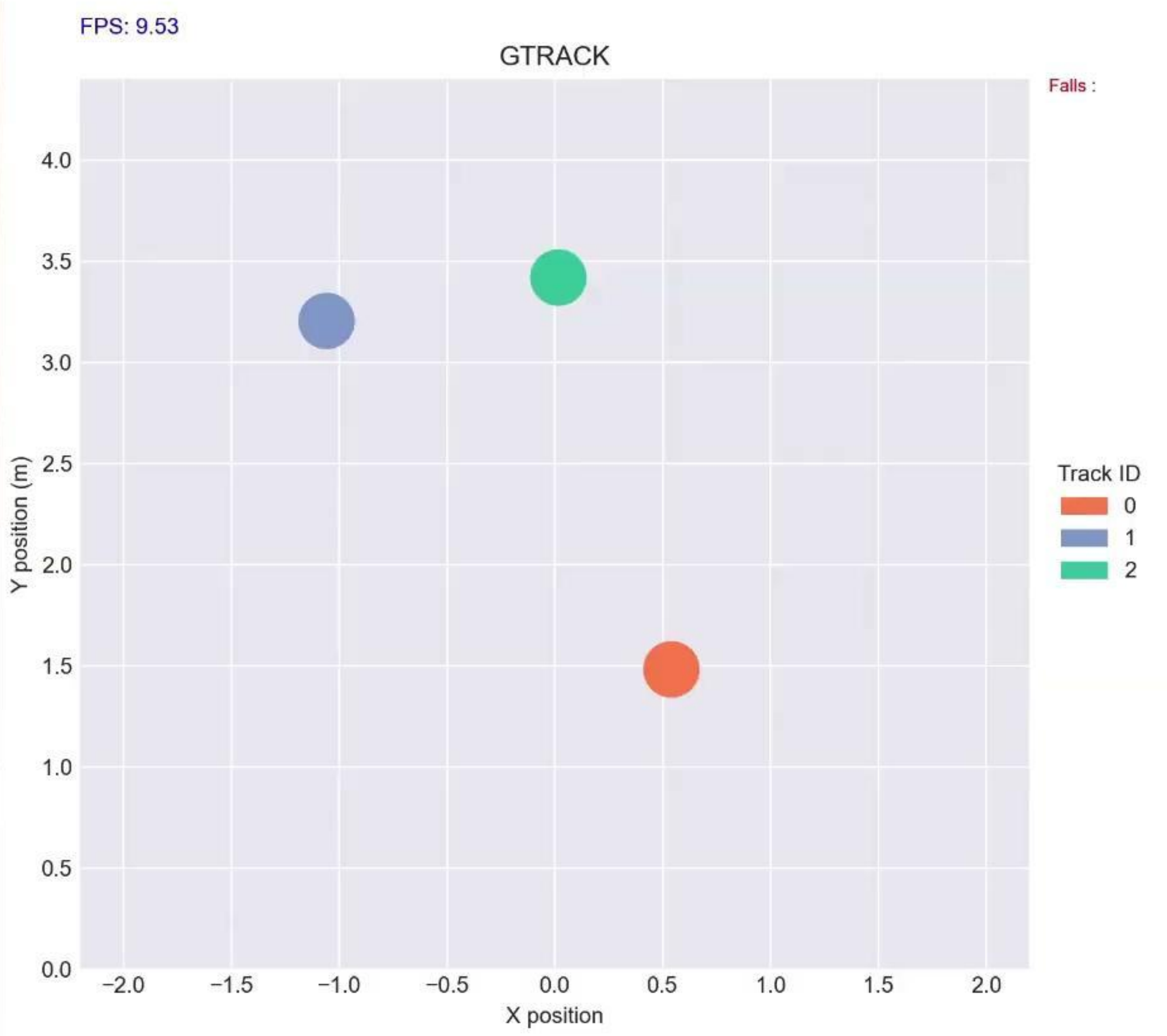




FPS: 11.41
GTRACK
Falls :
Track ID
Y position (m)
X position
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0



FPS: 10.03
GTRACK
Falls :
Y position (m)
X position
Track ID
0
1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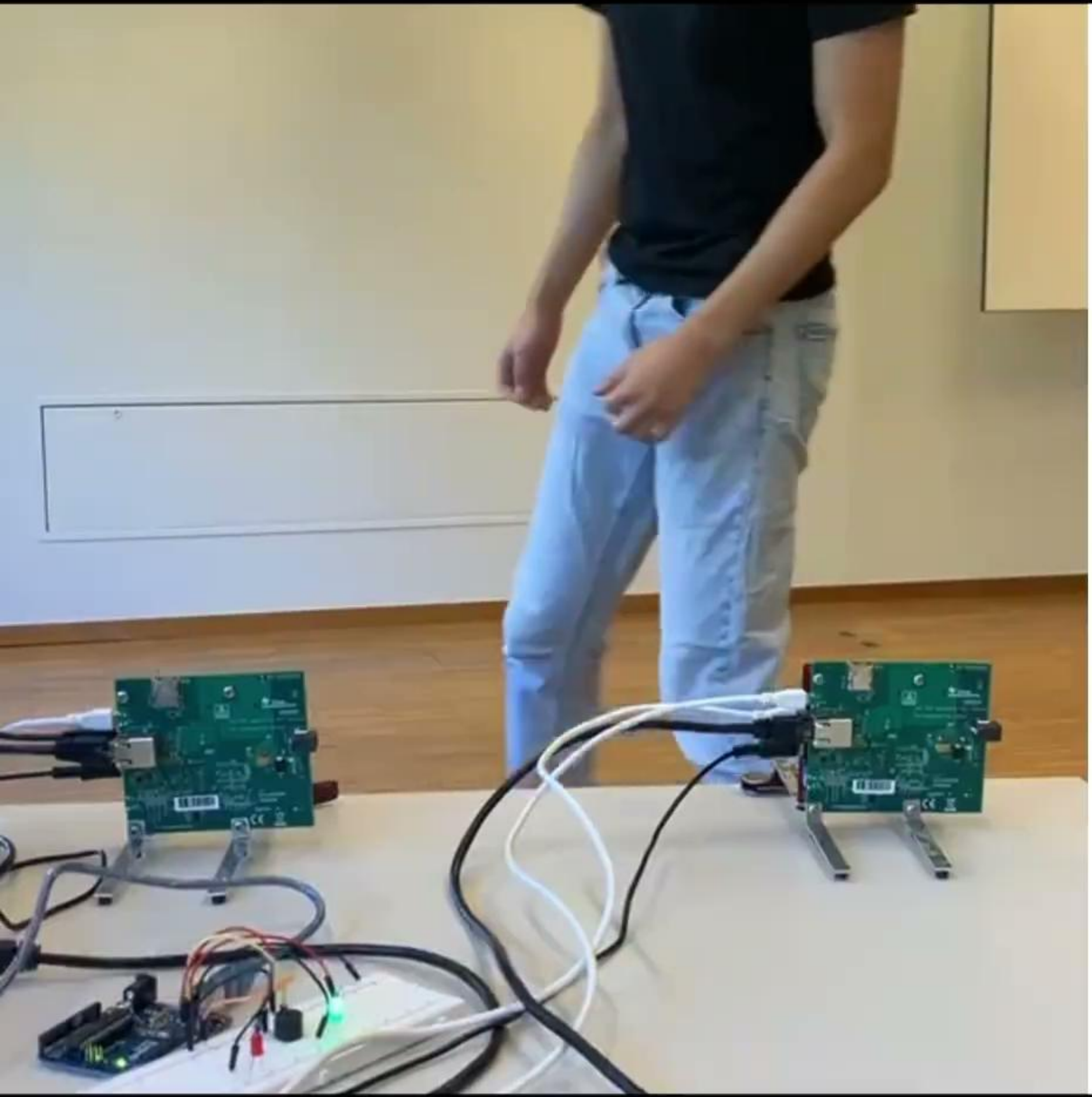
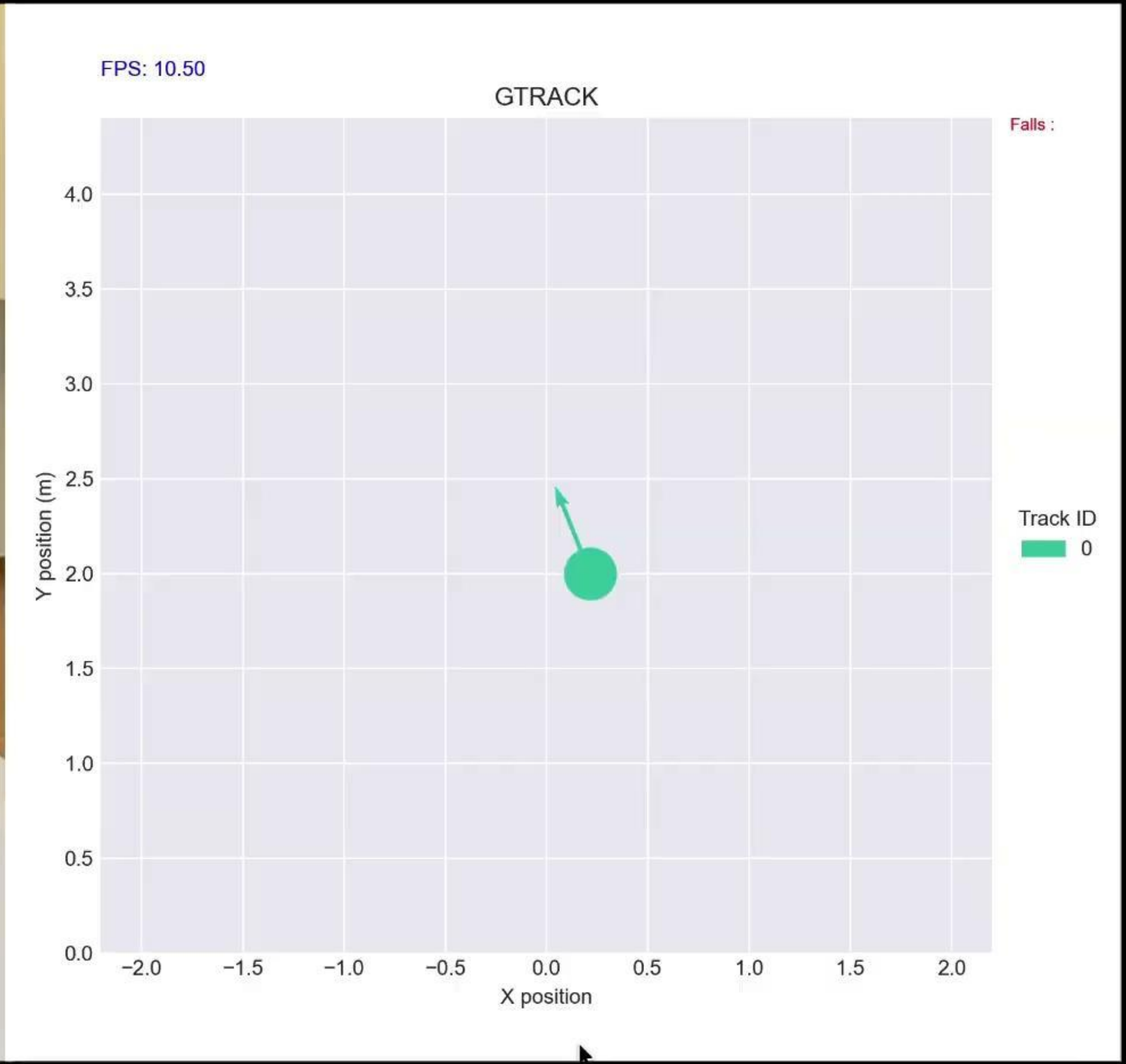
GTRACK
Falls :
Y position (m)
X position
Track ID
0
1
2
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0

Fall Detection

Presentation




FPS: 10.50
GTRACK
Falls :
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Y position (m)
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
X position
Track ID
0

Question Time